Year 11 Mathematics

Algebra 8

12.1 Simultaneous Equations

Simultaneous Equations

STARTER · 5 MIN

Date: _____ **Title:** Simultaneous Equations

LEARNING OBJECTIVES

- Solve simultaneous equations by multiplying both equations
- Set up and solve simultaneous equations from problems
- Solve a linear and a quadratic equation algebraically

STARTER *Answer in your book. No calculators.*

1. Solve $3x = 21$
2. $2x + y = 10$ and $x = 3$. Find y.
3. Multiply $2x + 3y = 7$ by 2
4. Expand $(x + 3)^2$

Starter — Answers

STARTER

Tick or correct in green pen

1. $x = 21 \div 3 = \mathbf{7}$

2. $6 + y = 10, \quad y = \mathbf{4}$

3. $4x + 6y = \mathbf{14}$

4. $\mathbf{x^2 + 6x + 9}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can solve simultaneous equations by multiplying both equations	☐
SILVER	I can set up and solve simultaneous equations from problems	☐
GOLD	I can solve a linear and a quadratic equation algebraically	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Simultaneous Equations

INTRODUCTION · 3 MIN

Two equations with two unknowns: find the values that make both true.

Elimination

Make one letter's coefficients match, then add or subtract.

Same signs: subtract

Substitution

Rearrange one equation and put it into the other.

$y = 2x - 1$

Forming equations

Choose letters and write two equations.

Let a = adult price

Non-linear

Substitute the linear equation into the quadratic.

Solve the quadratic

Examples 1 and 2 — Multiplying Both Equations

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint make the number in front of one letter the same

Example 1

Solve $3x + 2y = 16$ and $2x - 3y = -11$.

Example 2

Solve $5x + 2y = 11$ and $3x + 4y = 15$.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

×3: $9x + 6y = 48$

×2: $4x - 6y = -22$

Add: $13x = 26$

$x = 2$

$6 + 2y = 16$

$y = 5$

Tip: make the y terms match.

Example 2

×2: $10x + 4y = 22$

Subtract: $7x = 7$

$x = 1$

$5 + 2y = 11$

$y = 3$

Tip: sometimes only one equation needs multiplying.

Task 1

PRACTICE · 9 MIN

BRONZE *Solve each pair*

Hint make the number in front of one letter the same

1. $2x + 3y = 13$ and $3x + 2y = 12$
2. $4x + 3y = 18$ and $3x - 2y = 5$
3. $5x - 2y = 4$ and $3x + 5y = 21$
4. $3x + 4y = 5$ and $2x - 3y = 9$
5. $7x + 3y = 1$ and $2x + 5y = -8$

★ $6x - 5y = 1$ and $4x - 3y = 1$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $5y = 15, \quad y = 3, \quad x = \mathbf{2}$

2. $17x = 51, \quad x = 3, \quad y = \mathbf{2}$

3. $31x = 62, \quad x = 2, \quad y = \mathbf{3}$

4. $17x = 51, \quad x = 3, \quad y = \mathbf{-1}$

5. $29x = 29, \quad x = 1, \quad y = \mathbf{-2}$

★ $y = 1, \quad x = \mathbf{1}$

Examples 3 and 4 — Forming Simultaneous Equations

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint write two equations, then eliminate

Example 3

3 adult and 2 child tickets cost £34. 2 adult and 5 child tickets cost £41. Find the price of each.

Example 4

The sum of two numbers is 40. Their difference is 12. Find the numbers.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$3a + 2c = 34$$
$$2a + 5c = 41$$
$$11c = 55$$
$$c = 5$$
$$3a + 10 = 34$$
$$a = 8$$

Adult £8 and child £5

Tip: define your letters first.

Example 4

$$x + y = 40$$
$$x - y = 12$$
$$2x = 52$$
$$x = 26$$
$$\mathbf{y = 14}$$

Tip: add the equations.

Task 2

PRACTICE · 9 MIN

SILVER *Write two equations, then solve*

Hint write two equations, then eliminate

1. Two numbers add to 50 and differ by 8. Find them.
2. 4 pens and 3 rulers cost £5.10. 2 pens and 5 rulers cost £5. Find each price.
3. 5 teas and 2 cakes cost £13. 3 teas and 4 cakes cost £13.40. Find each price.
4. $y = mx + c$ passes through (1, 5) and (3, 11). Find m and c.
5. Ali and Sara's ages add to 30. In 6 years Ali will be twice as old as Sara. Find their ages now.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $2x = 58$, **29 and 21**

2. $7r = 490$, **pen 75p and ruler 70p**

3. $7t = 12.60$, **tea £1.80 and cake £2**

4. $2m = 6$, $m = 3$, $c =$ **2**

5. $a - 2s = 6$, $3s = 24$, **Sara 8 and Ali 22**

Examples 5 and 6 — One Linear, One Quadratic

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint substitute the linear equation into the other

Example 5 (4 marks)

Solve $y = 2x + 1$ and $y = x^2 - 2$.

Example 6 (4 marks)

Solve $x + y = 5$ and $xy = 6$.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$x^2 - 2 = 2x + 1$$

$$x^2 - 2x - 3 = 0$$

$$(x - 3)(x + 1) = 0$$

(3, 7) and (−1, −1)

Tip: find y for each value of x.

Example 6

$$y = 5 - x$$

$$x(5 - x) = 6$$

$$x^2 - 5x + 6 = 0$$

(2, 3) and (3, 2)

Tip: rearrange the linear equation first.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint substitute the linear equation into the other

1. Solve $y = x + 4$ and $y = x^2 - 2$ (4 marks)

2. Solve $y = 3x - 2$ and $y = x^2$ (4 marks)

3. Solve $x + y = 7$ and $xy = 12$ (4 marks)

4. Solve $y = 2x$ and $x^2 + y^2 = 45$ (4 marks)

5. Solve $y = x - 1$ and $x^2 + y^2 = 25$ (5 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $x^2 - x - 6 = 0$, **(3, 7) and (−2, 2)**

2. $x^2 - 3x + 2 = 0$, **(1, 1) and (2, 4)**

3. $x^2 - 7x + 12 = 0$, **(3, 4) and (4, 3)**

4. $5x^2 = 45$, **(3, 6) and (−3, −6)**

5. $x^2 - x - 12 = 0$, **(4, 3) and (−3, −4)**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Solve $2x + y = 9$ and $x - y = 3$

SILVER Two numbers add to 20 and differ by 6. Find them.

GOLD Solve $y = x^2$ and $y = x + 6$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $3x = 12, \quad x = 4, \quad y = \mathbf{1}$

SILVER $2x = 26, \mathbf{13 \text{ and } 7}$

GOLD $x^2 - x - 6 = 0, \mathbf{(3,\ 9) \text{ and } (-2,\ 4)}$

Exam Practice — Multiplying Both Equations

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint make the number in front of one letter the same

Question 1

(3 marks)

Solve the simultaneous equations

$3x + 2y = 12$

$2x + 3y = 13$

x = ... y = ...

Question 2

(3 marks)

Solve the simultaneous equations

$4x + 3y = 11$

$3x - 2y = 4$

x = ... y = ...

Exam Practice — Forming Simultaneous Equations

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint write two equations, then eliminate

Question 3

(4 marks)

3 burgers and 2 drinks cost £13.70

2 burgers and 3 drinks cost £11.30

Work out the cost of one burger and one drink.

..... burger £..... drink £.....

Question 4

(3 marks)

The line $y = mx + c$ passes through (2, 7) and (5, 16).

Find m and c.

m = ... c = ...

Exam Practice — One Linear, One Quadratic

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint substitute the linear equation into the other

Question 5

(4 marks)

Solve the simultaneous equations

$y = x^2 - 3$

$y = 2x$

.....

Question 6

(4 marks)

Solve the simultaneous equations

$x + y = 6$ $xy = 8$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

× 3: 9x + 6y = 36 — M1

× 2: 4x + 6y = 26

subtract: 5x = 10, x = 2 — A1

6 + 2y = 12, y = 3 — A1

Tip: check in the second equation.

Question 2

× 2: 8x + 6y = 22 — M1

× 3: 9x − 6y = 12

add: 17x = 34, x = 2 — A1

8 + 3y = 11, y = 1 — A1

Tip: +6y and −6y: add.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

6b + 4d = 27.40 — M1

6b + 9d = 33.90

subtract: 5d = 6.50 — M1

drink £1.30, burger £3.70 — A1 A1

Tip: write two equations first.

Question 4

7 = 2m + c, 16 = 5m + c — M1

subtract: 3m = 9, m = 3 — A1

7 = 6 + c, c = 1 — A1

Tip: substitute each point into y = mx + c.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$x^2 - 3 = 2x$ — M1

$x^2 - 2x - 3 = 0$ — M1

$(x - 3)(x + 1) = 0$

$x = 3, y = 6$ — A1

$x = -1, y = -2$ — A1

Tip: rearrange to = 0 before factorising.

Question 6

$y = 6 - x,\ \ x(6 - x) = 8$ — M1

$x^2 - 6x + 8 = 0$ — M1

$(x - 2)(x - 4) = 0$

$x = 2, y = 4$ — A1

$x = 4, y = 2$ — A1

Tip: make y the subject of the linear equation.